Heterogeneous initial states: weak instruments and independent-record inference

S16. Heterogeneous initial states and independent weak instruments

Consider independent records $Y_r = x_{\theta_0,u_r} + \eta_r$ of a deterministic polynomial ODE. The initial states may form a deterministic triangular array; alternatively they may be iid draws from a compact set, with the following conditions imposed conditionally on them. The sensor covariance is C_ν. Each Hermite-corrected, exactly integrated weak moment obeys

$$E\{m_r(\theta_0, \nu_0) \mid u_r\} = 0, \qquad m_r(\theta, \nu) = b_r - G_r(\nu)\theta.$$

This identity follows by taking expectations of the corrected polynomial and applying the ODE weak identity separately to each trajectory. Its validity is uniform over allowed initial states; the distribution of u_r need not be estimated. The numerical version needs the quadrature condition below.

S16.1. Numbered assumptions and fixed-noise estimating equations

(A1) The coefficient vector has fixed dimension p and belongs to an open identifiable neighborhood. Trajectories and their first coefficient/initial-state sensitivities exist on $[0, T]$. Initial states lie in a compact set. For some $\delta > 0$, the instrumented derivative matrices B_r and scores ψ_r defined below have uniformly bounded $4+\delta$ moments, conditional on the independent training data. Matrix norms are Frobenius norms. These are bounds on the products, rather than separate second-moment bounds on instruments and residuals.

(A2) Records and any auxiliary randomizations are independent across r. Noise is Gaussian with known positive covariance C_{ν_0}. Weak tests and quadrature rules are fixed independently of each record's noise. The coefficient structure is supplied.

(A3) For every record there are two conditionally independent views Y_r^+ and Y_r^- with mean x_{θ_0,u_r} and covariance $2C_{\nu_0}$. An instrument $L_r^\pm$ is measurable with respect to its own view and independent auxiliary training data. Corrected moments $m_r^\pm$ use the variance of their own view. The instrument map is fixed conditional on that training data and has the moment bounds required in (A1).

(A4) Put $B_r = (L_r^+ G_r^- + L_r^- G_r^+)/2$, $a_r = (L_r^+ b_r^- + L_r^- b_r^+)/2$ and $\psi_r = a_r - B_r\theta_0$. Conditional on the training data, $n^{-1}\sum EB_r \to B$ with B nonsingular, and $n^{-1}\sum \mathrm{Var}(\psi_r) \to V_\psi$ positive definite. For a deterministic initial-state array these are limits of array averages, not assumptions of identical record distributions.

(A5) Exact integration is used, or the instrumented numerical score means $\mu_{r,h}(q) = E\psi_{r,h}(q)$ satisfy $n^{-1}\sum_r \sup_{q\in Q} \|\mu_{r,h}(q)\| = o(n^{-1/2})$ for the compact instrument-pilot set Q. The moment

bounds and limits in (A1) and (A4) apply to the numerical scores as well. With external training, the bound is conditional on that training. This uniform bound controls the mean contribution even for a random complement pilot. A fixed integration step can leave a nonzero defect and then fails this centered continuous-parameter condition. An $o(1)$ bound suffices for consistency alone.

Theorem 3 (heterogeneous-record weak estimating equations). With known sensor covariance C_{ν_0} and under (A1)–(A5), conditional on the independent training data,

$$\hat{\theta}_n = \left(\sum_{r=1}^{n} B_r\right)^{-1} \sum_{r=1}^{n} a_r \xrightarrow{P} \theta_0, \qquad \sqrt{n}(\hat{\theta}_n - \theta_0) \Rightarrow N(0, B^{-1} V_\psi B^{-\top}).$$

The empirical covariance of the p-dimensional record scores, combined with the mean derivative $\bar{B}_n$, consistently estimates this sandwich. Thus a rich weak family requires a $p \times p$ empirical score covariance, while its within-record weights can come from the full model covariance in Proposition 3. Fixed-noise consistency applies to the stated independent-instrument construction. It does not follow merely by inserting a same-record pilot into $D_r^\top \Omega_r^{-1}$.

Proof. Conditional on u_r and training data, L_r^+ is independent of $m_r^-(\theta_0, \nu_0)$, and conversely. Hence

$$E(\psi_r \mid u_r, \text{training}) = \tfrac{1}{2} E L_r^+ \, E m_r^- + \tfrac{1}{2} E L_r^- \, E m_r^+ = 0.$$

By independence and (A1), $E\|n^{-1}\sum_r (B_r - EB_r)\|^2 \le C/n$, so (A4) gives $\bar{B}_n \to_P B$. Likewise $\bar{\psi}_n \to_P 0$. For any fixed vector c, the Lyapunov numerator for the centered sum of $c^\top \psi_r$ is at most Cn, whereas its variance to power $1 + \delta/2$ has order $n^{1+\delta/2}$. The ratio tends to zero. The scalar Lyapunov CLT and Cramer–Wold therefore yield $n^{-1/2}\sum_r (\psi_r - E\psi_r) \Rightarrow N(0, V_\psi)$. Assumption (A5) makes $n^{-1/2}\sum_r E\psi_r = o(1)$. The exact identity $\hat{\theta}_n - \theta_0 = \bar{B}_n^{-1}\bar{\psi}_n$ and Slutsky prove both limits.

For the sandwich, put $\widehat{\psi}_r = a_r - B_r \hat{\theta}_n$. Independence and the fourth-moment bound give $n^{-1}\sum_r \{\psi_r \psi_r^\top - E(\psi_r \psi_r^\top)\} \to_P 0$. Exact mean zero, or (A5) with bounded means, implies that its limit is V_ψ. Writing $d = \hat{\theta}_n - \theta_0 = O_P(n^{-1/2})$, Cauchy–Schwarz bounds the change caused by substituting $\widehat{\psi}_r = \psi_r - B_r d$ by

$$2\|d\| \left(n^{-1}\sum_r \|B_r\|^2\right)^{1/2} \left(n^{-1}\sum_r \|\psi_r\|^2\right)^{1/2} + \|d\|^2 n^{-1}\sum_r \|B_r\|^2 = o_P(1).$$

The fitted score mean is zero by the pooled estimating equation. Thus its empirical centered covariance, with optional factor $n/(n-1)$, converges to V_ψ; multiplication by $\bar{B}_n^{-1}$ on both sides gives the consistent covariance of the scaled estimator. This direct argument accommodates independent nonidentical records. In the iid case, the corresponding GMM results are Newey–McFadden Theorems 2.6 (consistency), 3.4 (normality) and 4.5 (variance estimation); their iid uniform laws are not invoked for arbitrary deterministic arrays [1, 2]. □

S16.2. Gaussian construction and local information reference

For known C_{ν_0}, draw $\zeta_r \sim N(0, C_{\nu_0})$ independently and set $Y_r^{\pm} = Y_r \pm \zeta_r$. Their cross-covariance is $C_{\nu_0} - C_{\nu_0} = 0$; jointly Gaussian views are independent, have identical mean and covariance $2C_{\nu_0}$. Together they recover the original record. This Gaussian randomization is an established data-fission construction [3]. It is used here to check finite-noise instruments, not proposed as a new general splitting method.

For a supplied clean trajectory, write $D_r = EG_r$ and $\Omega_r = \mathrm{Cov}(m_r(\theta_0, \nu_0))$. The deterministic optimal instrument within the prescribed linear moments is $L_r = D_r^\top \Omega_r^{-1}$. Direct calculation gives the oracle covariance

$$\left(\sum_r D_r^\top \Omega_r^{-1} D_r \right)^{-1}.$$

This is a fixed-family moment reference at finite noise. Proposition 3 evaluates Ω_r including all Hermite orders. A noisy $G_r^\top$ in place of $D_r^\top$ introduces $E(G_r^\top m_r)$ and can invalidate the zero-mean score. Independent instruments preserve the mean identity while generally changing the finite-noise covariance from this oracle expression.

In the local Gaussian experiment, each record has coefficient sensitivity S_r and initial sensitivity Z_r. The joint information for $\theta, u_1, \ldots, u_n$ has separate nuisance blocks. Its coefficient Schur complement is exactly

$$\mathcal{I}_{e,n} = \sum_{r=1}^{n} \left(\mathcal{I}_{\theta\theta}(u_r) - \mathcal{I}_{\theta u}(u_r) \mathcal{I}_{uu}(u_r)^{-1} \mathcal{I}_{u\theta}(u_r) \right) = \sum_r \mathcal{I}_e(u_r).$$

Theorem 2 applies record by record. With known shape and noise covariance $\epsilon^2 C$, a fixed test span gives the order-one moment information $D_r^\top \Omega_{r,1}^{-1} D_r$. Dense spans approach $\mathcal{I}_e(u_r)$ for any fixed finite collection of regular trajectories. Thus $(\sum_r \mathcal{I}_e(u_r))^{-1}$ is the sharp local reference for boundary-free influences in the product experiment. This exact block calculation alone supplies neither a fixed-noise incidental-parameter MLE theorem nor a growing-record, growing-span uniform efficiency result.

The symmetric independent-view score also preserves the leading influence under the following additional local conditions: its instruments converge in $L^{2+\delta}$ to the deterministic correct-covariance instruments; their perturbation is $O_P(\epsilon)$; weak moments have an L^2 expansion with $O(\epsilon^2)$ remainder; and covariance solves remain regular for the fixed span. Then

$$\tfrac{1}{2}\{L_r^+ m_r^- + L_r^- m_r^+\} = L_{r,0}\, m_{r,1}(\eta_r) + O_{L^2}(\epsilon^2).$$

Indeed the first-order moments are linear in $\eta_r \pm \zeta_r$, their average cancels ζ_r, and instrument perturbations multiply moments of order ϵ. The factor two in each view's covariance scales $L_{r,0}$ uniformly and cancels from the normalized estimating equation. Combined with Theorem 2, this proves the conditional local product-experiment efficiency statement for fixed n and sequential span enrichment. The fixed-noise CLT above and the local information reference describe complementary properties of the same weak scores; joint rates remain open.

S16.3. Covariance estimation and common-trajectory sufficiency

For an independently calibrated covariance estimate $\hat{\nu}$, asymptotic linearity of the averaged score adds $H\xi$ to its influence, where $H = E\partial_\nu\psi$ and ξ is the calibration influence with the appropriate relative sample-size scaling. The Newey–McFadden nuisance argument then gives its sandwich [2]. For estimated-covariance Gaussian fission, calibration error also perturbs cross-view independence; its contribution must be included or be negligible at the inference scale. The known-covariance construction proves the stated result without presuming those extra conditions.

Uncorrected residual-profile estimates from separate fixed-horizon trajectory fits need not be consistent as records accumulate: each fit estimates a new initial state with nonvanishing error. Even in a linear Gaussian model, fitting a mean projects part of the error out of the residual and changes its expected quadratic form. A residual estimator therefore requires a demonstrated bias correction or a separate consistent calibration scheme. Pair differences eliminate the mean only for equal trajectories. These facts prevent transferring the common-trajectory pair estimator to heterogeneous initial states.

If all records instead share one initial state and known covariance, the Gaussian path average is sufficient for the mean parameter: the sample contrasts are parameter-free independent Gaussian observations. Applying a single-record procedure to the path average reduces covariance to C/n. For a nonlinear polynomial, a Hermite covariance term of order q then scales as n^{-q}, whereas averaging separate corrected moments retains its fixed-amplitude contribution divided by n. Path averaging is the natural reference in this common-trajectory experiment. The old fixed-family GMM calculation is retained as Lemma S14.1 and a calibration diagnostic, rather than as an efficiency contribution.

S16.4. Cross-fitting across independent records

Partition the record indices into K deterministic folds I_k, fixed before observing the data, with K fixed and $|I_k|/n \to \pi_k \in (0,1)$. Fit a coefficient pilot $\hat{q}_{-k}$ using only records outside I_k. For $r \in I_k$, set

$$\psi_r(q) = \tfrac{1}{2}\{L_r^+(q)m_r^-(\theta_0,\nu_0) + L_r^-(q)m_r^+(\theta_0,\nu_0)\}, \qquad L_r^\pm(q) = L(q, Y_r^\pm).$$

Here the sensor covariance remains known. The instrument uses its own view, while the residual uses the independent opposite view. Conditional on the complement of fold k and on the initial states in I_k,

$$E\{\psi_r(\hat{q}_{-k}) \mid (Y_s^+, Y_s^-)_{s\notin I_k}, u_r\} = 0.$$

The population score mean is zero for every admissible coefficient pilot, including an imperfect one. This exact identity concerns the instrument pilot; covariance plug-in in Hermite corrections requires additional analysis. Cross-fitting is an established way to reuse records while preserving the training/inference separation [4].

(A6) For every fold, $\hat{q}_{-k} \to_P q_*$ for a deterministic q_* in the compact set Q; pilots are restricted to Q. Uniformly in r and deterministic $q, q' \in Q$, both maps $F_r = \psi_r$ and $F_r = B_r$ satisfy $E\|F_r(q) - F_r(q')\|^2 \le C\|q - q'\|^2$. The envelopes $\sup_{q\in Q}\|F_r(q)\|$ have uniformly bounded $4+\delta$ moments. Condition (A4) holds at q_*, and (A5) holds uniformly on Q. These conditions concern the implemented instrument map, including any bounded interpolation or clipping.

Corollary 2 (record-wise cross-fitting). Under (A1), (A2), (A4)–(A6), using the independent views in (A3) and replacing auxiliary training data by each fold complement, the pooled cross-fitted estimator has the same first-order distribution and consistent record-score sandwich as the deterministic instrument map at q_*. Every record enters the pooled estimating equation once; no external pilot sample is required.

Proof. Write $\Delta_{r,k} = \psi_r(\hat{q}_{-k}) - \psi_r(q_*)$ for $r \in I_k$, and condition only on the complement of this fold. The held-out terms are conditionally independent. With numerical integration their means may be nonzero, so first subtract their conditional means. The centered conditional second moment gives

$$E\left[\left\| n^{-1/2} \sum_{r \in I_k} (\Delta_{r,k} - E[\Delta_{r,k} \mid \mathcal{F}_{-k}]) \right\|^2 \; middle \mid \mathcal{F}_{-k}\right] \leq C \frac{|I_k|}{n} \|\hat{q}_{-k} - q_*\|^2.$$

Here $\mathcal{F}_{-k}$ is generated by the complement views. For a fixed tolerance ε, conditional Chebyshev on $\|\hat{q}_{-k} - q_*\| \leq t$ bounds the exceedance probability by Ct^2/ε^2; pilot consistency bounds its complement. Letting $n \to \infty$ and then $t \downarrow 0$ proves an $o_P(1)$ centered difference. The conditional mean sum has norm bounded by $2n^{-1/2} \sum_r \sup_{q \in Q} \|\mu_{r,h}(q)\| = o(1)$ by (A5). Summing the fixed number of folds therefore gives an $o_P(1)$ difference of normalized scores.

For the derivative, conditional Cauchy–Schwarz bounds $n^{-1} \sum_{r \in I_k} E[\|B_r(\hat{q}_{-k}) - B_r(q_*)\| \mid \mathcal{F}_{-k}]$ by $C(|I_k|/n)\|\hat{q}_{-k} - q_*\|$. Conditional Markov and the same event decomposition imply an $o_P(1)$ derivative difference. Conditional Markov applied to the squared score differences also gives $n^{-1} \sum_r \|\psi_r(\hat{q}_{-k(r)}) - \psi_r(q_*)\|^2 = o_P(1)$. The uniform envelopes make the average squared scores and derivatives $O_P(1)$. Cauchy–Schwarz then makes the score outer-product difference $o_P(1)$. Theorem 3 at the deterministic map q_* supplies its CLT and variance limit; the pooled linear identity and Slutsky give the cross-fitted limit. Finally, substituting the fitted coefficient in the scores uses precisely the preceding sandwich bound. Each conditional argument uses one fold complement separately; conditioning on all fitted pilots together would generally destroy held-out independence. □

The limiting pilot may be a pseudo-true point of the pilot procedure: exact score centering preserves consistency of the target estimator, while instrument quality determines its limiting variance. Consistency of the pilot suffices here because its score-mean effect vanishes identically; no generic $n^{-1/4}$ rate is asserted for other nuisance parameters. Fixed-span variance and local information efficiency remain distinct claims.

For finite-sample calibration, resample physical records within their original folds and refit every complement pilot in every bootstrap draw. Keeping copies in the original fold excludes every copy of a held-out physical record from its complement. Compute studentized marginal pivots and a quadratic joint pivot from each refitted estimate and its record-score sandwich. This procedure propagates pilot uncertainty that a frozen-instrument bootstrap omits. Its coverage is assessed numerically; the asymptotic result alone supplies no finite-sample 95% guarantee. Bootstrap validity additionally requires regularity of the pilot and estimating-equation functionals and an appropriate within-fold bootstrap limit; arbitrary heterogeneous deterministic arrays need separate justification. The simulation uses iid initial-state draws and balanced folds.

References

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